

Credit Name: CSE2920 - CSE Project C

Assignment Name: Buttons And LEDs

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to modify the given code to turn the LED off while the button is pressed, make the green button control the red LED and the red button control the green LED, and count the total number of button presses. Explanations of how I modified the code are below:

CODING:

1.

```
//Add Phidgets Library | You added a file called phidget22 wh
import com.phidget22.*;

public class ButtonsAndLEDs {
    //Handle Exceptions | Exceptions will happen in your code
    public static void main(String[] args) throws Exception{

        //Create | Create objects for your buttons and LEDs.
        DigitalInput redButton = new DigitalInput();
        DigitalOutput redLED = new DigitalOutput();
        DigitalInput greenButton = new DigitalInput();
        DigitalOutput greenLED = new DigitalOutput();

        //Address | Address your four objects which lets your
        redButton.setHubPort(0);
        redButton.setIsHubPortDevice(true);
        redLED.setHubPort(1);
        redLED.setIsHubPortDevice(true);
        greenButton.setHubPort(5);
        greenButton.setIsHubPortDevice(true);
        greenLED.setHubPort(4);
        greenLED.setIsHubPortDevice(true);

        //Open | Connect your program to your physical devices
        redButton.open(1000);
        redLED.open(1000);
        greenButton.open(1000);
        greenLED.open(1000);
    }
}
```

Phidgets library is imported

greenButton, greenLED, redButton, and redLED objects were created to interact with the physical phidgets box.

The HubPort number was set based on the appropriate object.

IsHubPortDevice set to true for all objects (confirming to the program that the objects are part of phidgets box).

Opened all of the objects, establishing a connection to the physical Phidgets box.

2.

```
int buttonPress = 0;

//Use your Phidgets | This code will turn on t
while(true){

    if(greenButton.getState()) {
        buttonPress++;
        System.out.println(buttonPress);
        Thread.sleep(50);
    }
    if(redButton.getState()) {
        buttonPress++;
        System.out.println(buttonPress);
        Thread.sleep(50);
    }
}
```

Int button press to count the number of button presses

While loop is used so that the following code keeps running.

If the green button or red button are pressed, every press results in the buttonpress variable increasing in value by 1. Sleep time of 50 ms so that the button presses are accurately registered.

3.

```
    }

    if( redButton.getState()){
        greenLED.setState(false);
    } else {
        greenLED.setState(true);
    }

    if(greenButton.getState()){
        redLED.setState(false);
    } else {
        redLED.setState(true);
    }

    Thread.sleep(50);
}
```

If the red button is pressed, the green LED turns off. Else , the green LED stays on.

If the green button is pressed, the red LED turns off. Else, the red LED stays on.

Sleep time of 50ms.

End of Program!